Kinova Gen3 Robotic Arm Simulation & Gripper Design

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Background of Microsurgery

- Extreme precision requirements – errors of even a few microns can cause complications.

- Limitation of human hand – fatigue and tremors reduce accuracy.

- Development of microsurgical robotics – enhanced stability and repeatability.

- Examples: Da Vinci and Symani systems.

Motivation for the Study

- Limitations of commercial systems (closed-source, costly, and not modular).
- Need for modularity – enables customization for different surgical needs.
- Low-cost requirement – improves accessibility.
- Goal: Enhance Kinova Gen3 gripper for delicate microsurgical tasks.

Robotic Platform Used

- Kinova Gen3 robotic arm (7 DOF)

- Equipped with Robotiq 2F-85 parallel gripper

- Controlled via ROS + MoveIt + Kinova Web App

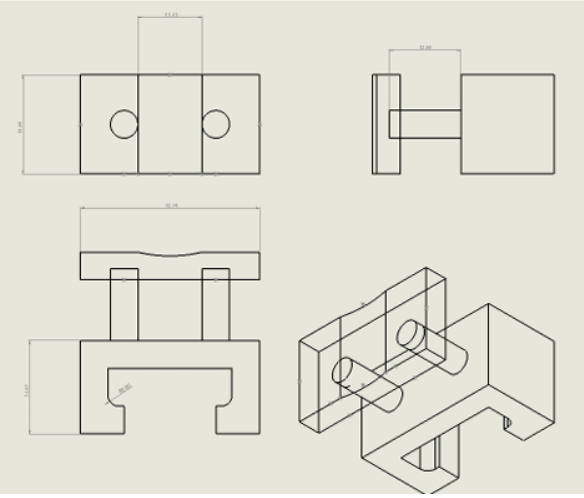
- Suitable for high-precision microsurgical manipulation

System Architecture Overview

- System Integration (ROS + MoveIt)
- Simulation (Gazebo + RViz)
- Physical Testing (Remote Control)

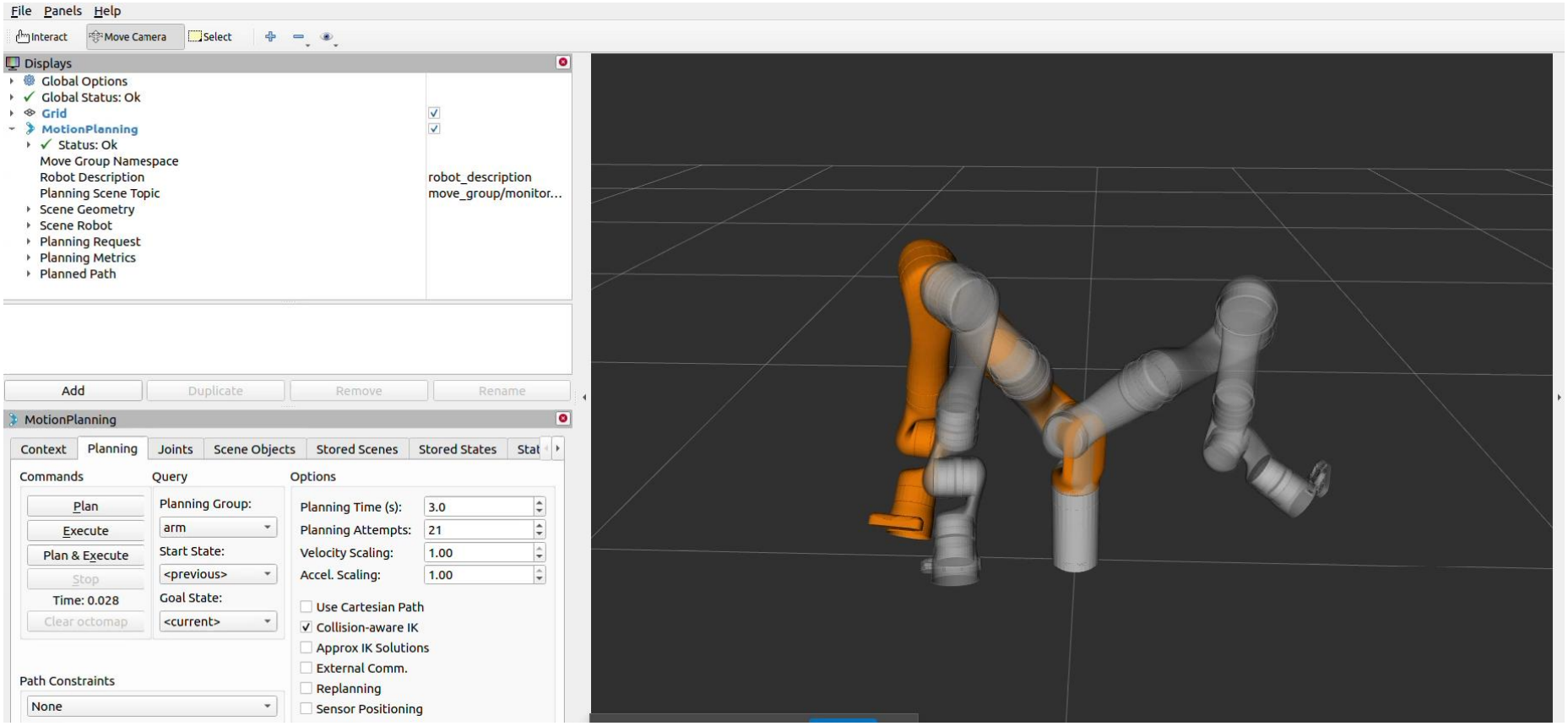
Custom Gripper Fixture – Key Points

- Designed in SolidWorks with dimensions based on official geometry
- 3D printed in TPU for flexibility and high friction
- U-slot ensures alignment; rounded edges prevent tool damage
- • Mounted on Robotiq gripper for real-world testing



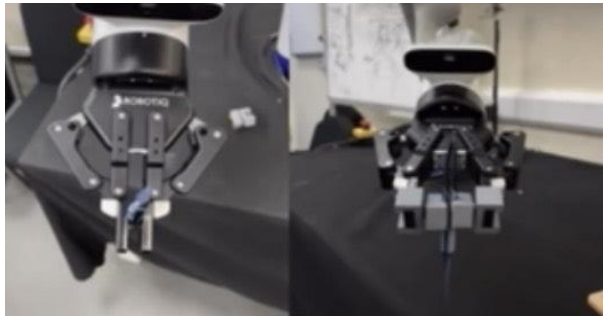
Simulation Results of Kinova Gen3

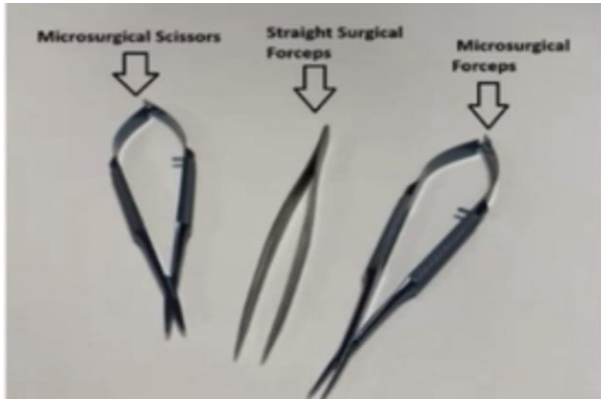
- 1. Joint Angle Limit Check – Verified compliance with set limits.
- 2. Reachability Verification – Demonstrated accurate reach to target positions.
- 3. Collision Detection – No self-collision detected during motion planning.



Methodology – Real Testing

- Remote control via Kinova Web App
- Surgical tools used: microscissors, forceps, needle holder
- Original vs Customized gripper comparison in grasping experiments.





Results & Analysis – Gripper Performance

Success Rate:

70% (original) → 90% (customized)

Tool Orientation Accuracy:

50% → 100%

Execution Time:

15.4s → 15.2s

Improved stability with TPU fixture, identical trajectory duration.

Design Comparison Table

Feature	Original Gripper	Customized Gripper
Gripper Success Rate	70%	90%
Orientation Stability	50%	100%
Execution Time	15.4s	15.2s
Contact Area	Low	High
Slippage	Often	Rare

Conclusions

The customized TPU fixture significantly improved grip stability and tool orientation.

Execution time remained constant due to identical paths.

Demonstrated effective integration of simulation and real-world control for surgical use.